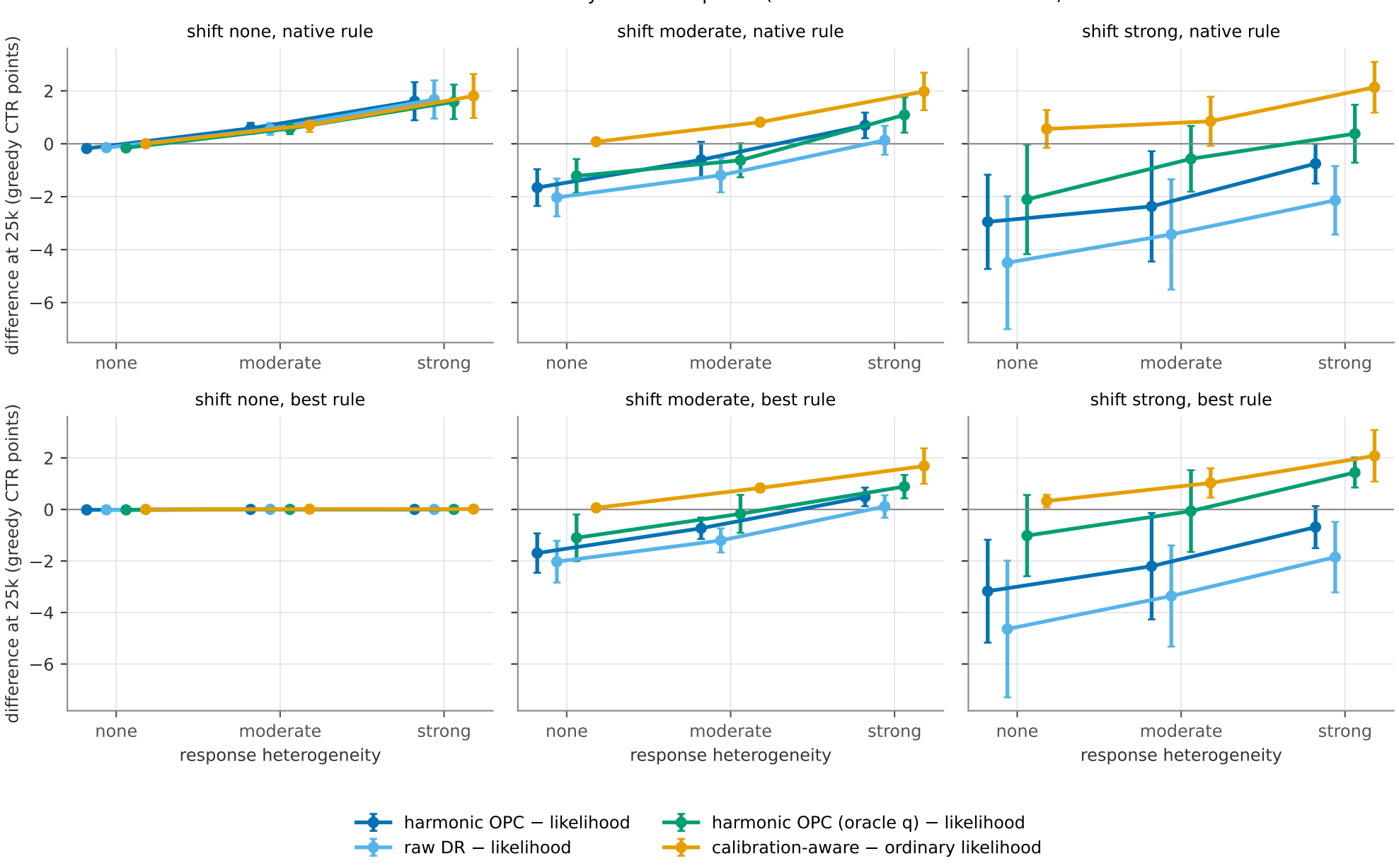
Paired differences at 25k by shift × response (mean and 95% CI over worlds)
shift none, native rule
shift moderate, native rule
shift strong, native rule
shift none, best rule
shift moderate, best rule
shift strong, best rule
difference at 25k (greedy CTR points)
2
0
−2
−4
−6
none
moderate
strong
response heterogeneity
harmonic OPC − likelihood
harmonic OPC (oracle q) − likelihood
raw DR − likelihood
calibration-aware − ordinary likelihood